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A Simulation Study of Photonic Nanojet Generated by a Combination of a Dielectric Microsphere and a Hemisphere Lens

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***Abstract*—Photonic nanojet (PNJ) is a highly focused light beam generated on the shadow side surface of a plane-wave-illuminated dielectric microcylinder, microsphere or hemisphere lens. Compound microsphere lenses allow image magnification and better resolution. Here, we report the PNJ generated by using a combination of a dielectric microsphere lens and a hemisphere lens. The numerical simulation results demonstrate that full width at half-maximum (FWHM) and working distance (WD) show a decreasing trend with the increases of the distance between the microsphere and the hemisphere. Compared with the hemisphere lens of smaller radius, the hemisphere lens with larger radius has greater FWHM and WD. The minimum FWHM is ~0.82λ and the corresponding WD is ~2λ.**

Keywords-photonic nanojet; microsphere lens; finite element method

I. INTRODUCTION

Photonic nanojets (PNJs) are of great interest due to their narrow beam waists and high intensities [1]. This phenomenon has been applied in many fields, such as enhanced backscattering, Raman scattering [2, 3], optical data storage [4], nanolithography [5], surface plasmon polaritons [6], all-optical switching [7], biomolecules/cells trapping and manipulation [8,9], detecting single nanoparticles [10], super-resolution imaging [11], biofluid monitoring [12], etc.

There are a few studies on composite lenses. By superimposing two microparticles in the vertical direction, magnification and field-of-view are expanded compared with a single microsphere [13-14]. Here, we report a structure that combines a dielectric microsphere with a hemisphere to expectantly promote the improvement of focusing performance. This work provides a numerical reference for the potential super-resolution imaging system based on a microsphere-hemisphere lens group.

II. RESULTS AND DISCUSSION

As shown in Fig. 1, the lens group is composed of a polystyrene microsphere (n_{ps}=1.59, on the top) and a silica hemisphere (n_{SiO2} = 1.46, on the bottom). Based on the finite element method (FEM), we study the PNJ characteristics of the microsphere-hemisphere lens group, i.e. FWHM and WD influenced by the size of the hemisphere, as well as the spacing between microsphere and hemisphere.

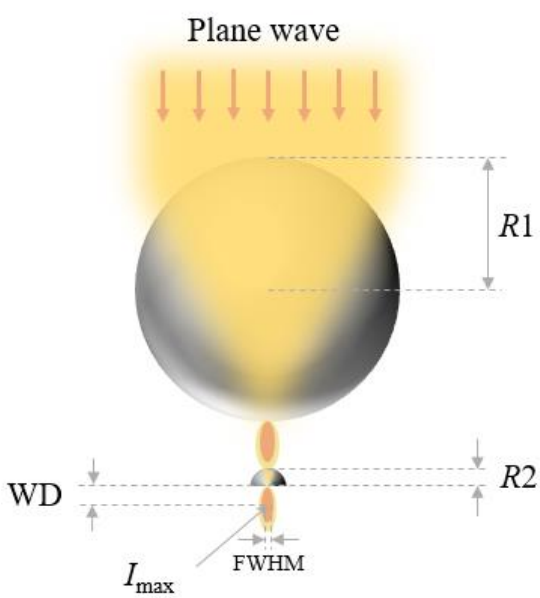

Figure 1. Schematic diagram of the microsphere-hemisphere lens group

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The incident light wave is set as a monochromatic nonpolarized plane wave, which travels from the top to the bottom with unit intensity. n_{ps}, n_{SiO2} and n_e, are the refractive indices of the polystyrene microsphere, the silica hemisphere and the environmental medium, respectively. $R1$ and $R2$ are the radius of polystyrene microsphere and silica hemisphere, respectively. d is the distance between microsphere and hemisphere, λ is the incident wavelength. FWHM is the full width at half-maximum of the central diffraction lobe as shown in the Fig. 1, WD is the distance from the bottom of hemisphere to the point of I_{max} (I_{max} is the point of maximum light intensity in the axial profile below the hemisphere). The center of the top microsphere is defined as the origin of the coordinate axis.

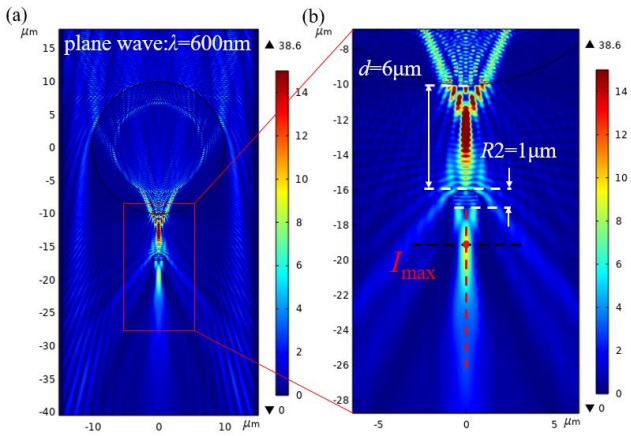

Figure 2. (a) FEM simulation results for a typical example of PNJ generated by a microsphere-hemisphere group with n_{ps} = 1.59, n_{SiO2} = 1.46, n_e = 1, $R1$ = 10 μm, $R2$ = 1 μm, d = 6 μm, and λ = 600 nm, I = 1. (b) Enlarged image of the rectangular area marked with a solid red line in (a).

The two-dimensional model shown in Fig. 2 corresponds to the schematic diagram of the microsphere-hemisphere lens group, which is used in the following numerical studies. We study two main PNJs parameters in this paper: FWHM and WD. Fig. 2. shows the light intensity distribution of the imaging system. The wavelength of the monochromatic nonpolarized plane wave is λ = 600 nm, and the light intensity is I = 1. The refractive indices of the polystyrene microsphere, silica hemisphere and environmental medium are n_{ps} = 1.59, n_{SiO2} = 1.46, and n_e = 1, respectively. The radius of the polystyrene microsphere $R1$ is 10 μm. The radius of the silica hemisphere ($R2$) ranges from 1 to 3 μm with step size of 1 μm. The distance between the microsphere and the hemisphere (d) is variable, ranges from 0 to 10 μm with steps of 0.5 μm. In this work, we show the parameters which have lobes on the shadow side of the hemisphere only.

As shown in Fig. 2, diffraction lobe is generated on the shadow side of polystyrene microspheres. After the incident light is focused, the silica hemisphere focuses the lobe again to form a new lobe on the shadow side of silica hemisphere.

The simulation results in Fig. 3 show that the FWHM, WD and the radius of the bottom hemispheres are related to PNJs. As the distance between microsphere and hemisphere increasing from 4.5 to 10 μm with step size of 0.5 μm, both FWHM and WD show a decreasing trend. The hemisphere lens with larger radius has greater FWHM and WD. The formation characteristics of PNJs are determined by the material properties and geometrical morphologies of the microparticle, the exterior environment, and the incident light. For the case of two microparticle lenses, the PNJ characteristics are not only determined by the above factors, but also influenced by the coupling and disturbance between the microsphere and the hemisphere, as well as the possible excited resonant whispering gallery modes [15].

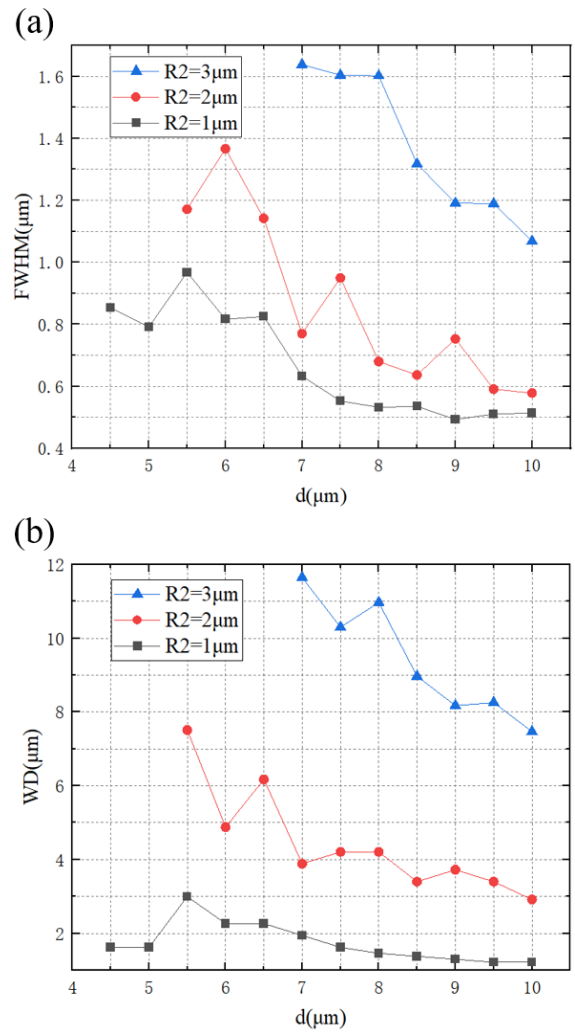

Figure 3. Variations of the two main PNJs parameters in microsphere and hemisphere lens group model. (a) FWHM and (b) WD as a function of different d.

III. Conclusion

Summarizing, we propose a model of a microsphere-hemisphere lens group. The numerical solution of the photonic nanojet in this model is provided by finite element simulation with the distance between microsphere and hemisphere increasing from 4.5 to 10 μm with step size of 0.5 μm, both of FWHM and WD shows a decreasing trend. The hemispherical lense with smaller radius has narrower

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FWHM and shorter WD. The minimum value of the FWHM is ~0.82λ and the corresponding WD is ~2.024λ.

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